EVERY QUASI-COMPACT HAUSDORFF SPACE HAS A CANONICAL MINIMAL EXTREMALLY DISCONNECTED COVER, UNIQUE UP TO ISOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the Stone-Cech compactification of a discrete space as extremally disconnected. Invoked inside the counted proof (source lines 5627-5632). Produces the quasi-compact Hausdorff extremally disconnected space covering the target.
- The existence of a minimal quasi-compact subset surjecting onto the target. Invoked inside the counted proof (source lines 5633-5635). Gives the minimal cover used for uniqueness.
- The lifting property of extremally disconnected quasi-compact Hausdorff spaces, and the statement that a surjection over an extremally disconnected target which is minimal is a homeomorphism. Invoked inside the counted proof (source lines 5636-5645 and 5650-5658). Produces the self-map and the isomorphism in the uniqueness argument.
- The statement that a non-identity surjective self-map of a Hausdorff space admits a proper closed subset covering the space with its image. Invoked inside the counted proof (source lines 5641-5643). Rules out a non-identity self-map of the minimal cover.
- The statement that a continuous map from a quasi-compact to a Hausdorff space is closed. Invoked inside the counted proof (source lines 5651-5652). Makes the comparison map closed.

Source reference. Original section example-stone-Cech-discrete: sources/stacks-topology/pinned-original-source.tex, source lines 5590–5590. Navigation pointer only: the counted text refers to the Stacks item example-stone-Cech-discrete, which is an existing labeled result of this same chapter and is not part of this card. The

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Source reference. Original lemma lemma-closed-map: sources/stacks-topology/pinned-original-source.tex, source lines 3474–3474. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-map, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-find-compact-subset: sources/stacks-topology/pinned-original-source.tex, source lines 5503–5503. Navigation pointer only: the counted text refers to the Stacks item lemma-find-compact-subset, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-isomorphism: sources/stacks-topology/pinned-original-source.tex, source lines 5481–5481. Navigation pointer only: the counted text refers to the Stacks item lemma-isomorphism, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-rainwater: sources/stacks-topology/pinned-original-source.tex, source lines 5573–5573. Navigation pointer only: the counted text refers to the Stacks item lemma-rainwater, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original proposition proposition-projective-in-category-hausdorff-qc: sources/stacks-topology/pinned-original-source.tex, source lines 5530–5530. Navigation pointer only: the counted text refers to the Stacks item proposition-projective-in-category-hausdorff-qc, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a quasi-compact Hausdorff space. There exists a continuous surjection $X' \to X$ with X' quasi-compact, Hausdorff, and extremally disconnected. If we require that every proper closed subset of X' does not map onto X, then X' is unique up to isomorphism.*

Proof. Let $Y = X$ but endowed with the discrete topology. Let $X' = \beta(Y)$. The continuous map $Y \to X$ factors as $Y \to X' \to X$. This proves the first statement of the lemma by Example example-stone-Cech-discrete.

By Lemma lemma-find-compact-subset we can find a quasi-compact subset $E \subset X'$ surjecting onto X such that no proper closed subset of E surjects onto X. Because X' is extremally disconnected there exists a continuous map $f : X' \to E$ over X (Proposition proposition-projective-in-category-hausdorff-qc). Composing f with the map $E \to X'$ gives a continuous selfmap $f|_E : E \to E$. Observe that $f|_E$ has to be surjective as otherwise the image would be a proper closed subset surjecting onto X. Hence $f|_E$ has to be id_E as otherwise Lemma lemma-rainwater shows that E isn't minimal. Thus the id_E factors through the extremally disconnected

space X'. A formal, categorical argument (using the characterization of Proposition proposition-projective-in-category-hausdorff-qc) shows that E is extremally disconnected.

To prove uniqueness, suppose we have a second $X'' \to X$ minimal cover. By the lifting property proven in Proposition proposition-projective-in-category-hausdorff-qc we can find a continuous map $g : X' \to X''$ over X. Observe that g is a closed map (Lemma lemma-closed-map). Hence $g(X') \subset X''$ is a closed subset surjecting onto X and we conclude $g(X') = X''$ by minimality of X''. On the other hand, if $E \subset X'$ is a proper closed subset, then $g(E) \neq X''$ as E does not map onto X by minimality of X'. By Lemma lemma-isomorphism we see that g is an isomorphism. $\square$